AI-Assisted Data Analytics Final Project

Marketing Campaign Response Prediction

Data Analysis ODC

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1. Dataset Overview

- **Dataset Description:** The `marketing_campaign.csv` dataset contains detailed information on customer demographics, purchasing behavior across various product categories (wines, fruits, meat, etc.), and interactions with previous marketing campaigns.
- **Business Domain:** Marketing and Customer Analytics.
- **Business Problem:** The company is experiencing low conversion rates on marketing campaigns (approximately 14.9% response rate). Marketing budgets are being wasted on targeting customers who are unlikely to accept the offers, leading to suboptimal Return on Investment (ROI).
- **Target Variable:** `Response` (Binary: 1 if the customer accepted the offer in the last campaign, 0 otherwise).
- **Project Objectives:** To analyze customer behavior, segment the audience, and build a predictive machine learning model that identifies customers most likely to respond to future campaigns, enabling targeted and cost-effective marketing strategies.

2. Data Preparation

- **Data Quality Issues:** The initial dataset contained missing values in the `Income` column and a few extreme outliers in customer age (e.g., birth years in the 1800s) and income.
- **Cleaning Process:** Missing values in `Income` were imputed using the median based on education and marital status groups. Extreme age outliers and unrealistic income values were removed to ensure model robustness.
- **Feature Engineering:** Several new, highly predictive features were engineered:
  - `Total_Spend`: Sum of all product spending categories.
  - `Total_Campaigns_Accepted`: The sum of campaigns 1 through 5 accepted by the customer.
  - `Total_Purchases`: Sum of deals, web, catalog, and store purchases.
  - `Total_Kids`: Sum of `Kidhome` and `Teenhome`.
  - `Tenure_Days`: Calculated from `Dt_Customer` to represent customer loyalty duration.
  - `Age`: Calculated using the year of birth.

  - Binary variables like `is_parent` and categorical simplifications for Marital Status (e.g., grouping 'Alone', 'Absurd', 'YOLO' into 'Single') and Education (e.g., 'Basic', '2n Cycle', 'Graduation', 'Master', 'PhD').

- **AI-Assisted Preprocessing:** Generative AI was utilized to suggest the most logical imputation strategies for missing income and to write optimized Python code for calculating date-based features (`Tenure_Days`).

3. Exploratory Data Analysis

- **Trends:** Only 14.9% of the customer base responded to the final campaign. However, response rates jumped to 80-90% for customers who had accepted 3 or 4 previous campaigns.
- **Relationships:** A strong positive correlation (0.79) was observed between `Income` and `Total_Spend`. Furthermore, `Total_Campaigns_Accepted` showed the strongest direct relationship with the target variable.
- **Outliers:** Identified and removed customers with abnormally high incomes (above $600,000) which skewed the spending distributions.
- **Important Observations:** Demographic segmentation revealed that PhD holders, singles, and high-income customers are notably more likely to respond to campaigns compared to married households with children.

4. Machine Learning

- **Models Implemented:** Logistic Regression, Random Forest Classifier, and Gradient Boosting.
- **Evaluation Metrics:** Due to the severe class imbalance (85% No, 15% Yes), Accuracy was discarded as the primary metric. Instead, the models were evaluated using **Precision**, **Recall**, and the **F1-Score**.
- **Model Comparison:** While tree-based models provided high accuracy on the majority class, they struggled to generalize the minority class without aggressive threshold tuning.
- **Best-Performing Model:** A **Class-Weighted Logistic Regression** with a custom decision threshold of **0.6**.
- **Justification:** This model achieved a Precision of 0.511, a Recall of 0.672, and an F1-Score of 0.581. By prioritizing Recall (67.2%), the business successfully identifies the majority of true responders. The threshold of 0.6 was chosen to strike an optimal balance, keeping false positives manageable while maximizing campaign reach.

5. Dashboard Summary

- **Dashboard Design:** The Power BI dashboard features a modern, professional dark theme with vibrant green and yellow accents. It is divided into two focused pages with persistent slicers on the left for seamless navigation.
- **KPIs:** Total Revenue, Total Customers, Average Customer Tenure, and Global Response Rate.

- **Visualizations:** Includes Donut charts for product category breakdowns, Bar charts for demographic sales analysis, and dynamic Tooltips (including Report Page Tooltips) for granular insights on hover.
- **Predictive Insights:** Page 2 (titled "Predictive Targeting Insights") acts as the analytical masterpiece. It visualizes the machine learning outputs, translating Feature Importance (e.g., the impact of `Recency` vs. `Total_Campaigns_Accepted`) into digestible business graphics.

6. KPI Explanation

1. **Overall Response Rate**
   - *Definition:* The percentage of total customers who accepted the final campaign.
   - *Target Value:* 20% or higher.
   - *Target Justification:* A 20% baseline indicates a healthy targeting strategy, ensuring marketing funds are spent efficiently.
   - *Performance:* Current performance is 14.9% (Underperforming), highlighting the strict necessity for the deployed ML model.
2. **Average Spend per Customer (AOV)**
   - *Definition:* Total revenue divided by the total number of purchasing customers.
   - *Target Value:* $650
   - *Target Justification:* Based on the 75th percentile of current spending, aiming to push average customers into the premium tier.
   - *Performance:* Varies heavily by demographic; PhD and high-income segments exceed this target, while lower-income segments drag the average down.

7. AI-Generated Insights Summary

- **Key Trends:**
  - Only 14.9% of customers responded to the last marketing campaign, indicating the current targeting approach reaches many uninterested customers.
  - Past campaign engagement is by far the strongest predictor of future response: customers who accepted 0 past campaigns responded at $\sim$9%, while those who accepted 3-4 past campaigns responded at 80-90%. Loyalty and repeat engagement compound strongly.
  - Education, marital status, and income all show meaningful patterns: PhD holders, single/divorced/widowed customers, and high-income customers all respond at notably higher rates than their counterparts.
  - `Total_Spend` and `Income` are highly correlated (0.79), suggesting spend behavior and income largely move together — useful for future customer segmentation, but a caution flag for models sensitive to multicollinearity.
- **Model Interpretation:**

  - The final model (Logistic Regression, class-weighted, threshold=0.6) achieves Precision=0.511, Recall=0.672, F1=0.581 — meaning it correctly identifies about 67% of true responders while keeping false alarms manageable.
  - Model coefficients confirm `Total_Campaigns_Accepted`, `Tenure_Days`, and being a PhD holder increase response likelihood, while `Recency` (days since last purchase) and being Married/Together decrease it.
  - Accuracy was intentionally not used as the primary metric, since it's misleading on an imbalanced target — a model predicting "No" for everyone would appear accurate (85%) but provide zero business value.
- **Business Implications:**
  - Repeat responders are a goldmine — this segment should be prioritized in every future campaign, since they're far likelier to convert again.
  - High-income and highly-educated segments show above-average responsiveness — a premium/tailored campaign track for these groups is likely to outperform blanket campaigns.
  - Married/cohabiting customers with children respond less — messaging or offers for this segment may need to be rethought (e.g., family-oriented promotions) rather than assuming a one-size-fits-all approach.
- **Actionable Recommendations:**
  1. Build a "loyalty" campaign track specifically targeting customers who accepted 1+ past campaigns — this group has the highest conversion potential.
  2. Segment high-income/high-education customers into a premium campaign path with tailored offers.
  3. Investigate why married/cohabiting households with kids underperform — consider testing different messaging or incentives for this segment before excluding them from campaigns entirely.
  4. Use the model's probability score (not just a hard yes/no) to rank customers by likelihood to respond, allowing marketing to prioritize outreach efficiently within budget constraints.

8. Final Conclusions

The current blanket-approach to marketing is inefficient, yielding only a 14.9% success rate. By shifting to a data-driven strategy leveraging our Class-Weighted Logistic Regression model, the marketing team can predict campaign responders with 67% recall.

The interactive Power BI dashboard successfully democratizes these complex ML findings, allowing stakeholders to identify key drivers like historical campaign acceptance and customer tenure. Implementing the targeted strategies recommended in this report will significantly reduce marketing overhead and maximize the conversion rates of future campaigns.